

Determinants of Box Office Revenue

A cross-country empirical comparison of the US and South Korea, 2020–2025

OLS regression two markets

646 films post-clean

Genre × budget interactions

MODEL

OLS

US R^2

0.38

KOREA R^2

0.44

MARKETS

2

A global shock, two very different recoveries

-70%

global box-office revenue in 2020
— the largest single-year drop in
the industry's history

COVID-19

United States

World's largest box-office economy

- Producer-driven model
- Powered by budgets, IP, marketing scale
- Slow reopening under capacity limits

South Korea

Among Asia's most open theatrical markets

- Exhibitor-driven model
- High screen density, heavy cinema-going
- Rebounded far faster than the US

OBJECTIVE

Quantify what drives theatrical revenue in each market

Parallel OLS models with genre dummies and budget interactions, so the two markets can be read side by side.

United States

$$\text{Revenue} = \beta_0 + \beta_1 \text{Budget} + \text{genres} + (\text{Budget} \times \text{genre}) + \varepsilon$$

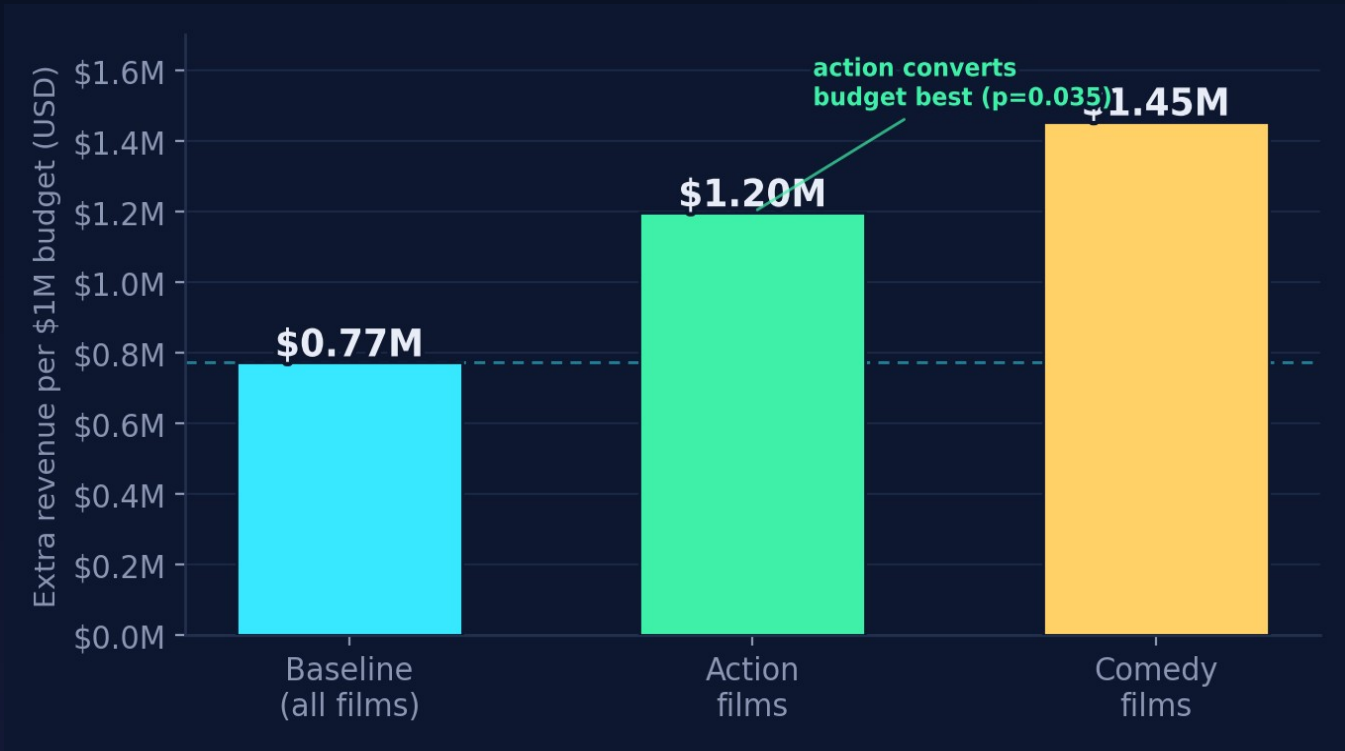
Dependent	Domestic box office (USD)
Key input	Actual production budget (US\$M)
Genres	Action · Drama · Comedy · Horror
Observations	323 films (post-clean)
Source	Box Office Mojo

South Korea

$$\text{Revenue} = \beta_0 + \beta_1 \text{Screens} + \text{genres} + (\text{Screens} \times \text{genre}) + \varepsilon$$

Dependent	Box office (KRW)
Key input	Screens × attendance (proxy)
Genres	Keyword-detected dummies
Observations	Foreign films, 2020–2025
Source	KOFIC / KOBIS

Budget is the engine — and action films convert it best



+\$771K

revenue per additional \$1M of budget (p<0.001)

+\$424K

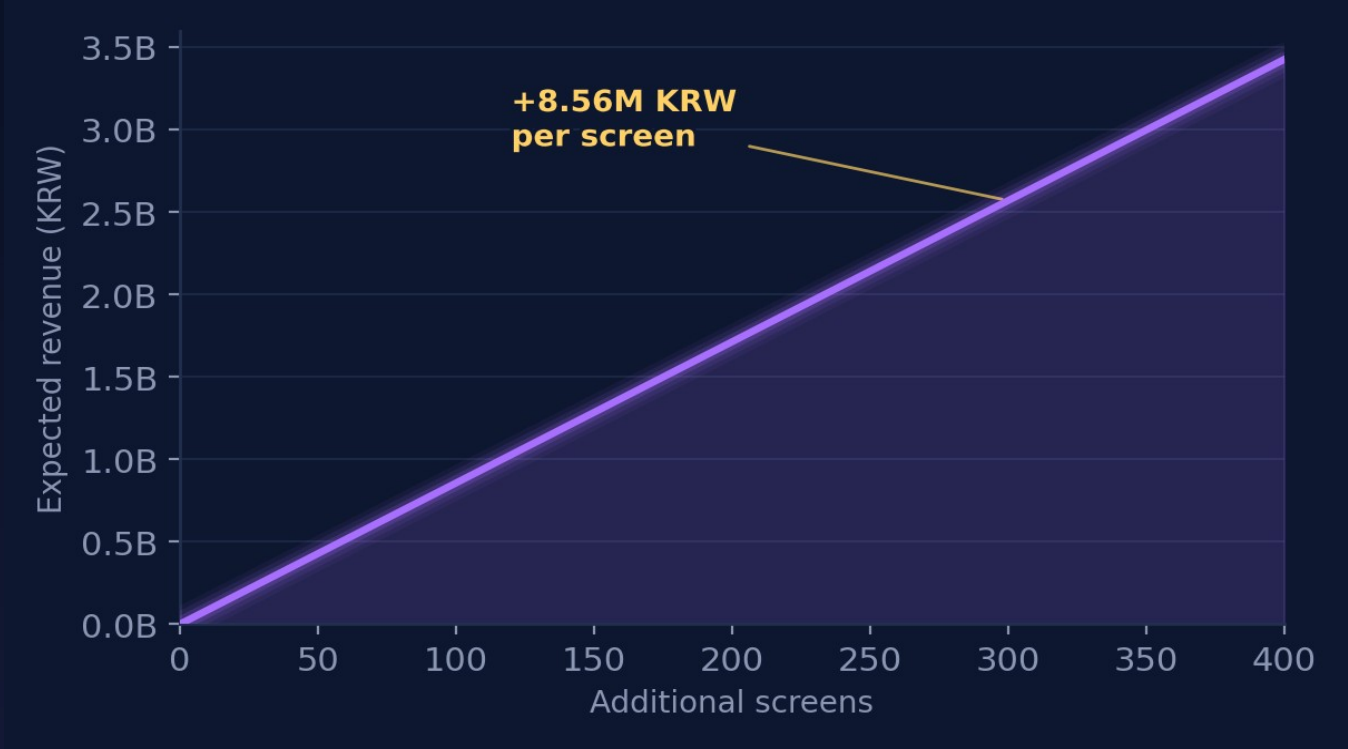
extra per \$1M for action films — budget × action is positive & significant (p=0.035)

0.38

R² — typical for cross-sectional film revenue models

Genre alone doesn't move revenue once budget is controlled for — production scale dominates.

Screens are the story — distribution reach beats genre



+8.56M

KRW of expected revenue per additional screen (t = 60.2, p<0.001)

0.44

R² — screens alone explain more than the full US budget model

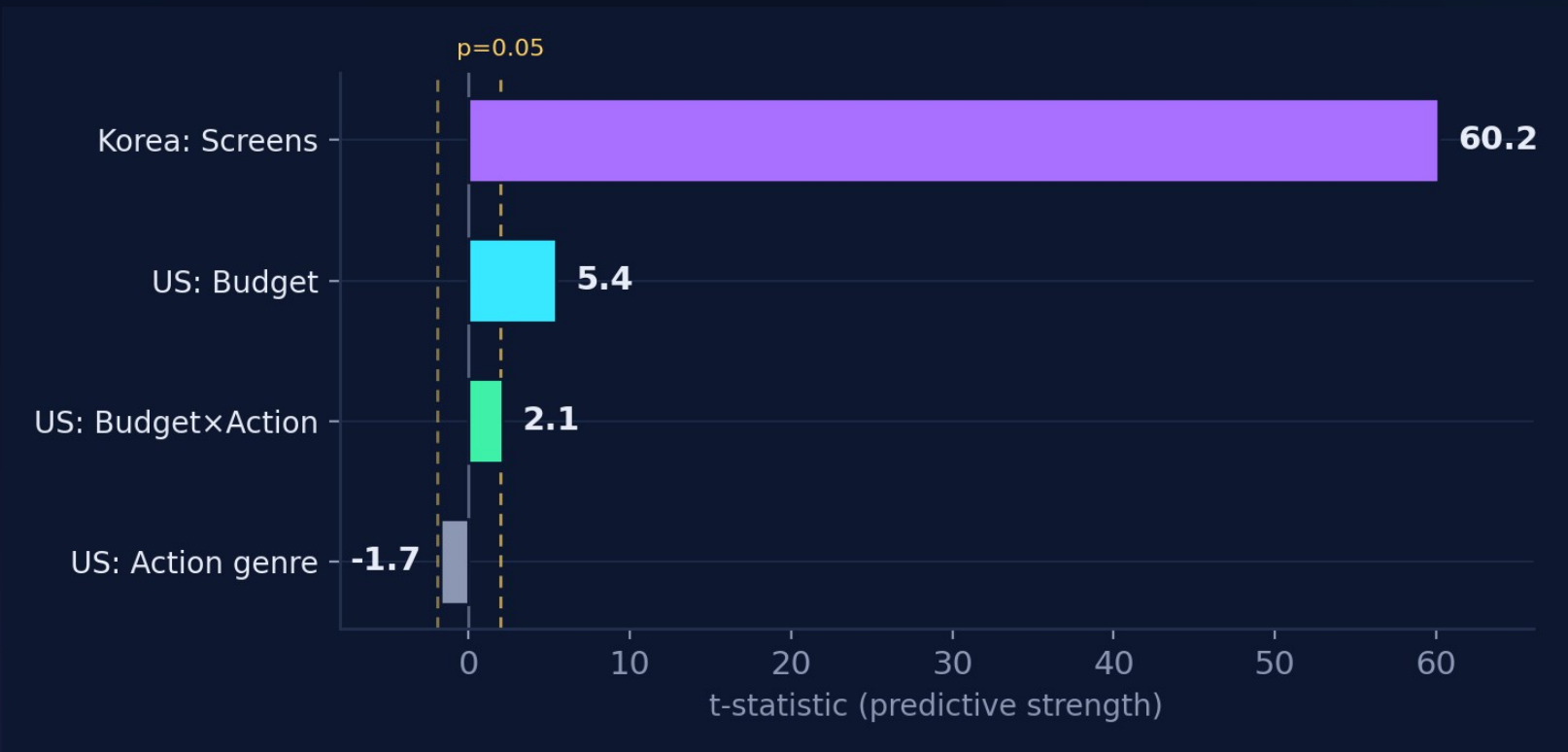
NA

most genre & interaction terms drop out — collinearity among foreign releases

Screen count proxies distributor confidence — access to screens is more decisive than genre-specific spend.

Different levers, measured on one scale

t-statistics across both models. Korea's screen effect is in a different league of statistical strength.



Why it matters

Screens (Korea) post a t of 60 — an almost mechanical link between exposure and revenue.

Budget (US) is strong but far more variable, reflecting genuine box-office uncertainty.

Only actionxbudget clears significance among US interactions.

Two structures, two mechanisms

Feature	United States	South Korea
Main predictor	Budget	Screens
Market structure	Producer-driven	Exhibitor-driven
Genre significance	Action matters	No significant effect
Interaction effects	Present	Not identifiable
R ²	0.38	0.44

Industrial structure — not universal factors — shapes what drives revenue in each market.

Honest about the proxy



Korea has no public budgets

KOBIS doesn't release production budgets, so investment scale is proxied by screens × attendance.



Possible endogeneity

The proxy is partly determined by box-office performance itself, which can inflate its correlation with revenue.



Multicollinearity

Foreign releases cluster in similar genres, collapsing genre and interaction terms — read those coefficients with caution.

The proxy doesn't invalidate the model — it highlights a real structural difference between the two datasets.

Success in cinema isn't universal — it's institutional

01

US: money talks

Production budget is the strongest, most consistent predictor — and action films convert marginal dollars best.

02

Korea: access talks

Screen count dominates. In a tightly coordinated exhibition system, distribution reach beats genre-specific spend.

03

Structure decides

The same global shock produced different revenue mechanics — driven by local institutions and information availability.

Data, methods & references

Method

- Ordinary least squares (OLS), estimated in R.
- US: revenue on production budget, four genre dummies, and budget×genre interactions.
- Korea: screens (× attendance proxy) replace budget; keyword-detected genre dummies mirror the US structure.
- US $n = 323$ films (post-clean); Korea = foreign releases, 2020–2025.

Selected references

- De Vany & Walls (1999) — uncertainty in the movie industry
- Elberse (2007) — the power of stars; returns to budget
- Hsu (2016) — returns to scale in blockbuster production
- Lee (2011); Oh & Park (2020) — Korean screen allocation
- Litman (1983) — predicting theatrical success
- Vogel (2020) — Entertainment Industry Economics

Sources: Box Office Mojo (US); Korean Film Council (KOFIC) / KOBIS (South Korea).